CalGames 2026 Content Desk: run the desk

The short guide for volunteers

Setting up? Start with QUICK-START.pdf (one page). This guide is the next step: which computer does what, what each screen says, and what to do when something looks wrong. The desk itself says most of this on its front page and in its **Doors check**, in plain words, at the moment it matters. The long reference is docs/handbook-reference.md; nobody needs it to run the show.

1 · The AV table: which computer does what

Several Windows computers sit side by side at the AV table. **Only one of them runs the Content Desk.** Every other computer either shows a desk page or runs something the desk only reads. If you are not sure which computer is in front of you, open the desk address in its browser: the first line of the front page says either **This computer runs the Content Desk.** or **This computer does not run the Content Desk.**, with the desk's address.

COMPUTER OR SCREEN	WHAT RUNS ON IT	HOW YOU CAN TELL	LEAVE IT ALONE?
The desk laptop (CalGames)	`CalGamesContentDesk.exe`, the Content Desk itself, and its desk console in the browser	Front page: *This computer runs the Content Desk.*	Keep it on, plugged in, lid open, on the field network. Closing the browser does not stop the show; never end the desk in Task Manager during a match
The graphics computer (its HDMI goes through a Teranex into the switcher's overlay input)	A browser, full screen, on `ADDRESS/s/program`	Black, with the desk's score graphics once a match loads; Cheesy Arena's own score on black if the desk stops	Leave it full screen. It is the overlay on the stream and the LED wall
The switcher computer (rack PC)	ATEM Software Control: cameras, transitions, and the key that lays the graphics over the cameras	The ATEM control window	The AV crew's. The desk only reads the switcher and cannot change it
The multiview monitor	The ATEM's own picture of every input	A grid of camera pictures	Nothing to do
The mixer computer	X Air Edit: microphones, music and the stream's sound	The X Air Edit window	The AV crew's. Keep the music channel out of the mix that feeds the stream
The field computer	Cheesy Arena, the scoring system	Match play screens	The scorekeeper's and the FTA's. The desk only reads it
The music machine (the laptop wired to the hall's speakers)	A browser on `ADDRESS/s/house`	The House audio player page	Leave it open; it plays the walk-ups and clips the desk fires
Projector or LED wall, venue TVs, pit TVs	A browser, full screen, on a **Pick a screen** card	The picture itself	Nothing to do; they reconnect by themselves
Announcer tablet, Judge Advisor's device	**Announcer tablet** and **Judge Advisor** from the front page	Their own pages	Theirs

The stream itself comes out of the switcher's program output through the Web Presenter; no computer runs it. **Put a strip of tape on each monitor** with its row name from this table. It takes two minutes and saves every question later.

Where the desk address is. On the desk laptop's front page, in **Start here**: *This desk on the event network*, for example `http://10.0.100.23:8720/`. Write it on the whiteboard with the **Desk PIN** shown under it. Every other computer types that address.

2 · Setting up, in order

Each step ends with what you should see. The desk finds the field and the switcher by itself; nothing needs a file edited or a flag typed.

1. **Desk laptop.** Power, and a network cable into the field network (the switch Cheesy Arena and the ATEM are on). Copy `CalGamesContentDesk.exe` onto it and double-click it. Nothing appears for up to a minute the first time. A blue *Windows protected your PC* box is normal: **More info**, **Run anyway**. When Windows asks about its firewall, tick every network type and press **Allow access**. *You should see* the browser open on the front page, whose first line says **This computer runs the Content Desk.**
2. **Whiteboard.** Write the address and the **Desk PIN** from **Start here**.
3. **Codes.** In **Start here**, click **Change the codes**. **Event settings** opens with the settings code. Under **Access codes**, change the desk PIN, the settings code and the awards code; give the awards code to the Judge Advisor only. A change signs out everyone using the old code. *You should see* no *Still on the shipped default* line in **Start here**.
4. **Graphics into the switcher.** On the graphics computer, open the address and click **Put the graphics into the video switcher**, put the window on the screen that goes to the switcher, and press **F11**. The switcher's key stays exactly as it was set up for Cheesy Arena's overlay: same input, same luma key on DSK 1. Key only this one input. *You should see* black, and the desk's graphics once a match loads. On the desk laptop's front page, **Screens** names that computer.
5. **Projector, LED wall and TVs.** On each computer that feeds a display, open the address, click **Put the show on a projector or TV**, click the card for that display (table below) and press **F11**.
6. **Field and switcher.** Nothing to do. The front page says *Field: searching* until the field answers, then *Field: connected*; **Switcher** names the ATEM once it is found. If either stays searching once the hardware is up, see chapter 6.
7. **Music.** On the music machine, open `ADDRESS/s/house` and press **Turn this machine on as the house player**.
8. **Doors check.** On the desk laptop, open **This laptop runs the show** (the desk console) and read **Doors check**. Every row that is not green says what to do.
9. **Posters, and a look at every screen.** In **Start here**, **print the QR posters** for the doors, the stands and the pits. Before any real match, *Fill the screens with sample data* (the link in **Start here**) puts invented teams on every screen so you can check each display; all of it says SAMPLE and it clears itself when a real match loads.

Pick a screen shows these cards, in this order:

FOR	SCREEN	ADDRESS	WHAT IT SHOWS
The projector, or the stream	**Program**	`/s/watch?screen=program`	The main show picture: who is playing, the live score, the final score
The TVs beside the field	**Side screen**	`/s/watch?screen=side`	Who is on deck and the current rankings, rotating
Pit TVs	**When do we play?**	`/s/watch?screen=next`	When each team plays next, with start times that follow the field
The trivia monitor	**Trivia**	`/s/watch?screen=trivia`	The crowd game, with its join code between questions
The arcade TV	**Arcade**	`/s/watch?screen=arcade`	Scores from the side tournament

3 · What the screens look like, and why

Black or white? `ADDRESS/s/program` opened by itself in a browser is graphics on black: right for the switcher's luma key, which drops the black out. The projector and TVs go through **Pick a screen**, which adds the purple CalGames backdrop. A page that is see-through (inside OBS, or with `?key=alpha` in its address) is painted white by a plain browser; the **Screens** line on the desk laptop's front page names any computer showing that and the address to open instead.

Cheesy's overlay as a backup. The graphics computer's page keeps Cheesy Arena's own score display loaded behind the desk's graphics. If the desk stops answering for about 3 seconds, or loses the field for 10 seconds while Cheesy still has it, that same input shows Cheesy's score instead, by itself, and goes back to the desk's graphics once both are answering again. Nobody presses anything, and the page never reloads while the desk is away.

Scores. Every number on air is solid white. When the field cannot send the score and the desk manager types it, the final screen says **Leading** (or **Level**) under the alliance instead of **Winner**, so a guess is never called the result. The announcer's tablet still says when a score is a guess.

RP chips in the score bar and on the final screen: a ranking point earned is a white chip in the alliance colour; one not yet earned is white on a dark plate.

What the projector says before a match loads, under CALGAMES 2026:

THE LINE, WORD FOR WORD	WHAT TO DO
Connecting to the desk at http://ADDRESS.	Wait a few seconds
No field connection. Matches can be run from the desk console.	The desk is not connected to the field yet. It finds the field by itself; meanwhile **Load a match by hand** works
Connected to the field. Waiting for it to load a match (its Test Match does not count).	Ask the scorekeeper to load a practice, qualification or playoff match
In yellow: *Cannot reach the desk at http://ADDRESS. Check that the desk computer is on and the Content Desk is running.*	The desk laptop is off or asleep, or this computer is on another network. It heals by itself
A second line, in yellow: *No picture from the field feed at HOST: check that the camera is on and its address is right. Until then the graphics show on this backdrop.*	Only with a field feed set on Event settings. Fix the camera or its address
The backdrop with no words	Somebody pressed **Blank**: press **Auto**

The score bar fills the bottom of the frame with its numbers centered. A TV that crops the edges of the picture (most LED walls and the stream do not) can open the program with `?safe=1` on the address to keep the text clear of the edge.

4 · Match day on the desk console

Open **This laptop runs the show** on the desk laptop and enter the PIN. The console opens in **Simple**, only what a match day needs; **Everything** shows every panel. With no OBS, the camera controls fold behind **Using OBS? Show the camera controls**. Every key works in both views.

The field drives the show. When a match loads, the program shows the alliance overview; the score bar comes in when the field arms; the clock and scores run live; the final score appears when the score posts and stays until the field's own audience screen moves on (about 20 seconds, never more than a minute). Leave it alone.

YOU WANT TO	USE
Put a stuck main screen back on the show	**Main screen (program)**: **Auto**
Empty the main screen	**Blank** (`B`)
Explain a pause	**Delay card**: **Field delay** and the others; **Hide card** takes it down
Tell the whole building something urgent	**Safety message**: **Put on every screen**
Run an awards ceremony	**Awards ceremony**: **Show the award**, then **Reveal the winner** when the announcer says the name
Mark a replay moment	`Space`
Show a name strip	Lower third: `T` shows it, `Y` hides it

If the field goes down, the console shows **Field down? Run matches by hand** by itself: **Load a match by hand**, then the match steps, which are Preview, Arm, Start, End, Post score, Abort (the keys 1 to 5, and 0 for Abort). Typed scores look the same on air; the final screen says **Leading**, not **Winner**. Hand scoring keys: `Q W E` for red, `A S D` for blue.

Holding a key does its action once. The console never takes a key while you are typing in a box.

5 · Event settings

`ADDRESS/s/setup`, with the settings code. Changes apply at once unless the page says otherwise.

- **Access codes**: the desk PIN (only from the desk laptop), the settings code and the awards code. The Judge Advisor can also change the awards code on their own page.
- **Connections**: **Field address (Cheesy Arena)**, **Video switcher address (ATEM)**, **the** FRC Nexus API key** (for the queue on the side screens), and which switcher input carries the desk's graphics. Press **Save connections**. Each box shows what the desk sees right now. Leave the addresses empty to let the desk look by itself.
- **Restart the desk** (desk laptop only): the desk stops and starts again in a few seconds. Screens reconnect by themselves; consoles ask for the code again.
- The event's content: names, sponsors, the run of show, slides.

6 · When something is wrong

Read the front page first: its status lines and **Something not working?** answer most things in a line or two, and the desk console's **Doors check** says what to do for every row that is not green.

YOU SEE	DO THIS
A white page	That computer has the see-through page. The **Screens** line on the desk laptop's front page names it and the address to open
Black on the graphics computer	Correct until a match loads
Field: searching that never ends	Check the desk laptop is on the field network (event Wi-Fi is usually a different network). Still searching: ask the FTA for the field's address and type it under **Field address (Cheesy Arena)**
Switcher says no switcher found	Fine on a setup without one. With the ATEM: type its address (four numbers, for example 10.0.100.40) under **Video switcher address (ATEM)**
Switcher says the switcher turned the desk down	It has no free connection: close any copy of ATEM Software Control nobody is using
Two score bars on the stream	Cheesy's overlay is keyed from its own computer as well. Key only the graphics computer's input
A TV cannot open the address	The firewall box from setup was missed, or the TV is on a Wi-Fi that keeps devices apart. Use the address in **Start here**, never `localhost`
Screens stop updating, or *reconnecting*	The desk stopped. Double-click the exe on the desk laptop again: the day comes back exactly as it was and every screen reconnects
A code does not work	Codes changed on Event settings sign out the old ones. The desk laptop's **Start here** shows the desk PIN
Doors check row **Event log** is red	The disk is full. Free space; do not restart the desk until it is green

The universal fix is double-clicking the exe on the desk laptop. If the desk is running it opens the front page; if it stopped, it starts again and rebuilds the day from its own log: scores, cards, the run of show.

The desk cannot start a match, abort a match, change a score, or change anything on the field or the switcher. It only reads them.

7 · The other jobs, in one paragraph each

Announcer. Announcer tablet: who is on the field, ranks, how close each alliance is to its ranking points, and how to say team names, all in words that can be read out.

Judge Advisor. Judge Advisor, with the awards code: enter each award's winner as judging wraps up. Winners stay off every screen and feed until the desk presses **Reveal the winner**.

Trivia. `ADDRESS/s/triviadesk`: one button whose label says what the next press does. The crowd joins from the QR code on the trivia screen.

Music. The music machine plays the walk-ups and clips the desk fires. Music never goes to the stream: the music machine is not wired to it, and on the mixer its channel stays out of the stream's mix.

Shout-outs. People in the stands send them from the QR posters; nothing reaches a screen until someone approves it on the desk console.

Appendix · Every address

ADDRESS	WHAT	WHO
`ADDRESS/`	Front page: what each computer is for, status, help	Everyone
`ADDRESS/s/desk`	Desk console	Desk laptop, PIN
`ADDRESS/s/program`	Graphics for the switcher, on black	Graphics computer
`ADDRESS/s/watch`	**Pick a screen** for projectors and TVs	Display computers
`ADDRESS/s/setup`	Event settings	Content lead, settings code
`ADDRESS/s/awards`	Awards	Judge Advisor, awards code
`ADDRESS/s/talent`	Announcer tablet	Emcees, PIN
`ADDRESS/s/house`	House audio player	Music machine
`ADDRESS/s/testcard`	Test card for checking a display	Whoever sets up a display
`ADDRESS/s/poster`	QR posters to print	Setup crew
`ADDRESS/s/next`, `/s/quiz`, `/s/gp`	When do we play, trivia, shout-outs	Phones in the stands